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Long-term Zooplankton Data Set with Precise Temporal and Spatial Resolution

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Abstract :

Long-term acoustic bathymetric data provide an opportunity to study the vertical distribution and behavior of zooplankton and other organisms. This article describes a small, low-cost, single-frequency (200 kHz) upward-looking water column profiler installed as part of the VENUS cable observatory in Saanich Inlet, British Columbia, Canada, to demonstrate depth sensing concepts. to introduce. Time cubes for inferring long time series of acoustic data. We describe the instrument calibration and calibrate the data to the absolute volumetric backscatter intensity. A time series of daily scattering measurements over two years is presented to illustrate the seasonal changes at this location. We also show that seasonal changes in the timing of diurnal vertical migration of zooplankton can be tracked with high temporal resolution using a simple algorithm and uncelebrated single-frequency data. As expected, Deal's vertical movement was found to be very closely related to solar declination, but a simple algorithm for determining timing could be confounded by fish or physical water movement. It happened often. More intelligent algorithms are needed that take advantage of the high temporal resolution of the data.

Keywords : Zooplankton, Bathymetric data

Introduction:

Shipborne sonar systems have been used to detect, map, and measure marine fish and zooplankton populations since the 1950s, and to map their distribution over relatively large areas.

It is routinely used for long resulting time sequences. Self-contained echo sounders, such as acoustic Doppler current profilers (ADCPs), are either moored at depth and pointing upwards, or attached to surface buoys and pointing downwards, producing a time series of acoustic backscatter, has been used for long-term vertical monitoring since the late 1990s. The 1980s used are the distribution and behavior of zooplankton and other populations in a single location [1]. However, ADCP is designed to measure current and is not optimized for backscatter measurements.

Currently, compact acoustic water column profilers (AWCP) have been specifically manufactured for backscatter measurements and more than 50 have been deployed worldwide for autonomous operation. These can operate at very high temporal and spatial resolution. Ping rates are up to 1 Hz and spatial resolution is up to 1/8 meter at depths up to 300 meters. This type of data can also be obtained from RSSI (received). However, AWCP offers significant performance benefits when considering long-term deployment and high sample rates. AWCP instruments are now also integrated into wired observatories, allowing near real-time interrogation and monitoring of the time-varying vertical distribution of scattered particles. Such instruments have been in continuous operation for more than four years at the VENUS cable observatory in Saanich Inlet, British Columbia.

Data obtained from mixed populations can also provide valuable information on the abundance and behavior of zooplankton populations over long periods of time. Additionally, it is often not possible to distinguish between echoes received from a small number of strong reflectors and echoes originating from a cloud of many small scatterers distributed throughout the beam. Dense clouds of eupausids and schools of large, low-density fish may look similar, but individual fish targets can often be identified by their behavior and the relatively short amount of time they spend in the river. AWCP's Time Varying Gain (TVG, described below) is set to the backscatter volume, so individual targets, such as individual fish, appear dimmer the further away they are. Although similar data can be obtained using ADCP, AWCP offers some important advantages in spatial and temporal resolution when considering long-term self-powered deployments. An ADCP operating at a depth of 150 m with a range resolution of 1 m and transmitting an ensemble of 30 pings per minute requires three times the battery capacity than an AWCP operating at the same depth and a range resolution of 0.5 m. The ping ensemble sends groups 5 times per minute. In most cases, operating the ADCP to obtain high-resolution backscatter data also degrades the performance of velocity measurements [2].

Materials and Methods:

Acoustic Water Column Profiler (AWCP) and Data:

The instrument, VENUS observatory, and research site have been previously described but are briefly summarized here. The inverting sounder is an ASL Environmental Sciences Inc. 200 kHz Acoustic Water Column ProfilerTM (AWCP). In autonomous operation, the instrument runs on internal batteries, but in the VENUS configuration, shore-based cables provide power and a true sounder. time communication. In this mode, there is no data storage as all data is immediately transferred via cable to the VENUS Data Management and Archive System "DMAS". Device ping rate is selectable. For shore power installations, a fast rate of 1 Hz is typical, but for long-duration battery-powered deployments, the rate can be lowered to save storage space [3].

On VENUS, the ping rate has been reduced to 0.3 Hz to avoid interference from other acoustic instruments on the same platform. The source level of the transmission is approximately 213 dB for 1 μPa at 1 m, and the length of the transmitted pulse can be selected between 4 and 1020 μs, although the Saanich Inlet instrument is nominally set to 300 μsec.

$$Sv = 10 \log (Ie/Ii)$$

Sv is measured in decibels (dB). Where is the intensity of the backscattered echo 1 m from the volume and Ii is the intensity of the incident sound? Since Ie is always less than Ii, Sv is always negative.

After transmission, the echo arriving at the converter passes through a preamplifier and a bandpass filter, then he passes through a TVG (Time Varying Gain) stage and an envelope detector to an analog-to-digital (A/D) converter. It will be sent. Amplitudes are sampled at 23.3 kHz and 8-bit resolution to a selected maximum range of 200 m or less. The digitization rate allows a range resolution of approximately 1/8 meter. The time-varying gain has a range of 80 dB and approximately follows the form 20logR + 2R. where is the absorption coefficient of 200 kHz sound in seawater and R is the range [5]. The TVG's starting gain can be selected from one of four values, allowing some adjustment to the limited dynamic range of currently installed 8-bit A/Ds. The count value N output by the A/D converter is from 0 to 255 and is related to the volumetric backscatter intensity Sv in region R and the sound source level SL by (1):

$$Sv = 20\log N - G(2R/c) - OCV - SL + 20\log R + 2\alpha R - 10\log(\tfrac{1}{2}c\tau\Psi)$$

Where c is the speed of sound, τ is the transmitted pulse width, Ψ is the equivalent solid angle of the transducer beam, G(2R/c) is the time-varying gain function of the instrument (dB), and OCV is the receive response. of the transducer. If the time-varying gain function of the instrument exactly matches the propagation and absorption losses, equation shows that the logarithm of N is directly proportional to Sv. The VENUS AWCP TVG is an analog circuit and is based on assumed sound speeds and absorption coefficients, which typically do not accurately balance propagation and absorption losses. To convert N to Sv, we need to correct for this difference.

The difference varies by range and can be more than 6 dB depending on the difference between the actual water properties and the water properties assumed by the TVG. By using measurements from TVG and the manufacturer's converter specifications, the deviation from direct proportionality is reduced to approximately 3 dB. Uncertainty can be further reduced by calibrating the overall instrument response using a known target. The Sv range that can be captured with AWCP is determined by the dynamic range of the A/D conversion, the selected TVG, and the internal noise level of the equipment. Increasing the overall gain lowers the maximum Sv at which the A/D saturates, emphasizing weak echoes, but also increasing instrument noise [6].

Lowering the overall gain allows stronger signals to be detected without saturation, but weaker signals are still below the A/D's minimum bit level. VENUS AWCP in Saanich Inlet uses TVG setting 2. It can detect echoes of approximately -84 to -40 dB Sv at a range of 50 m without saturation. At 100 m range, those limits are -73 to -36 dB.

Calibration of the Acoustic Water Column Profiler:

The combination of the electronic characteristics of the instrument and benchtop measurements of the specified characteristics of the transducer introduces some degree of uncertainty in the calibration of the instrument. This is because instrument calibration does not take into account variations in response that can occur when: The components of mold equipment work in combination rather than individually. This can be alleviated by measuring the response of a known target sphere in the water under controlled conditions, which allows the overall response of the instrument to be measured [7]. Calibration measures the difference (ΔTS) between the target strength measured in the field (TSin situ) and the theoretical value for a solid homogeneous elastic sphere estimated using a model (TSmodel). TSin situ was calculated by

$$TSin\ situ = 20\log N - G(2R/c) - OCV - SL + 40\log R + 2\alpha R$$

Here, TS is a measure of the reflectance of the target sphere. It is different from Sv because it is a standalone object. It depends on the volume. In February 2010, calibration was carried out on a buoy at the Ocean Technology Test Bed (OTTB), an underwater engineering laboratory operated by the University of Victoria in Saanich Inlet [1]. The AWCP transducer was mounted looking down from a depth of approximately 0.5 m, and data were recorded while the instrument observed a 38.1 mm diameter tungsten carbide sphere. The acoustic axis was found by varying the position of the sphere vertically and horizontally. Calibration measurements were performed using a sphere at seven different depths. CTD was used to measure temperature and salinity and calculate the speed of sound.

Calibration values were calculated using a sound velocity of 1476 m s-1. During the acoustic axis calibration run, received signals from 171 to 256 acoustic transmissions were averaged to determine the exact distance and backscatter intensity of his signals at each depth. TS model was estimated to be -39.49 dB based on the known parameters of tungsten carbide and the center frequency of AWCP. TS varied from -3.21 to -0.60 dB depending on the depth of the calibration sphere [8].

Acoustic volume increases with distance, so the greater the distance, the more likely the calibration sphere will be placed exactly on the acoustic axis. Therefore, we applied the observed TS at a maximum distance of 37 m as a constant correction for the system gain to calculate Sv (table 1).

Range (R) [m]	20logN [dB]	G(2R/c) [dB]	40logR+2αR [dB]	$TS_{in\ situ}$ [dB]	$\Delta TS = TS_{in\ situ} - TS_{model}$ [dB]
17.48	47.94	119.42	51.27	-42.21	-2.72
18.49	47.67	119.42	52.34	-41.41	-1.92
20.52	45.66	120.70	54.34	-42.70	-3.21
22.67	46.01	120.70	56.26	-40.43	-0.94
24.69	44.90	120.91	57.92	-40.09	-0.60
30.76	42.33	123.26	62.29	-40.64	-1.15
36.83	40.11	124.54	65.96	-40.47	-0.98

Table 1: Calibration values.

Results:

VENUS Observatory:

VENUS Cable Observatory is a permanent marine research facility that began operations in February 2006. Two cabled arrays are located on Canada's Pacific coast, operating in the coastal waters of southern Columbia, one of which is located in Saanich Inlet. Ground-based instrumentation platforms carry a variety of instruments, including CTDs, oxygen and dissolved gas sensors, permeability meters, and (in some cases) ADCPs and AWCPs. Other systems connected to the observatory include pan and tilt cameras and a hydrophone array to monitor ambient noise and marine mammal communication. Real-time data is transmitted via cable to a shore station and from there to the University of Victoria campus. Echograms showing data for the past 24 hours and other selected days (color-coded to display N in dB as a function of area and time) as an all-day chart. and as separate hourly plots. Past data is also available [9]. This article presents an example of 200 kHz data from a single-frequency AWCP installed at a depth of 96 m in Saanich inlet for the period June 2008 to May 2010. AWCP equipment has been operating continuously at this site since February 2006, with only brief interruptions due to maintenance or failures in other parts of the VENUS system (figure 1).

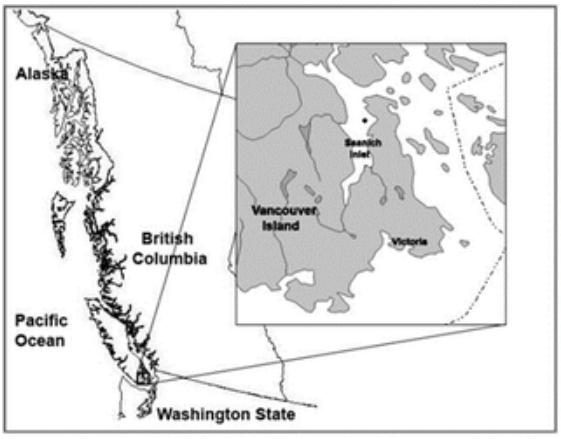

Figure 1: Saanich Inlet, on southern Vancouver Island, British Columbia, Canada.

Study location: Saanich Inlet, British Columbia Saanich Inlet is a temperate fjord in the coastal waters of southern British Columbia on Canada's Pacific coast. A shallow (75 m) sill at the north entrance of the inlet limits deep water exchange into the 230 m deep basin, and dissolved oxygen in inlet waters deeper than approximately 80 m is less than approximately 1 mL L-1. is. The upper extent of the anoxic (<0.1 mL O2 L-1) layer varies between 100 and 130 m, and deep water renewal usually occurs in autumn. Primary production in Saanich Inlet is high, with large phytoplankton outbreaks in the spring in April and May, and a few small flowers in the summer and autumn due to wind mixing [7].

The bay is home to a large population of Euphausia pacifica, which dominates a low-diversity macrozooplankton community and exhibits vertical diurnal migration (DVM) from the deep ocean to the surface. These relatively large animals cause a dramatic acoustic scattering layer in the 200 kHz VENUS AWCP profile. The bay is rich in Pacific herring (Clupea pallasi), Pacific hake (Merluccius productus), and walleye, which feed on both juveniles and adults.

In the AWCP data, both individual fish targets and schools of fish appear as high-intensity echoes with high temporal resolution. Their vertical movement and velocity through the sonar beam can be inferred from the temporal direction and horizontal length of the echo trajectories [10].

AWCP data is provided by the VENUS data management and archiving system as ASCII text or XML files. Text files were reformatted into binary format and ITT's commercial image processing software ENVITM was used for visualization and analysis. Five months of 8-bit AWCP data are sampled at 3 second time intervals and 13.2 cm vertical depth bins, producing files between 3.5 and gigabytes. Working with this large dataset can be difficult. For example, it is not possible to display the entire dataset at full resolution.We found it useful to think of AWCP data as a "cube" of [depth] x [time] x [day]. You can also imagine "cutting" 3D data. Cube". In different ways. Data is averaged over 1-minute intervals starting at 00:00 GMT (16:00 PST) each day. The first day of the displayed timeline is February 28, 2009, and the last day is July 31, 2009. For this presentation, the cube has been cropped to approximately 3m indicated by the hourglass shape at the top of the cube to eliminate surface returns and reveal calendar variations in the timing of rise and fall [8].

The columns of pixels on the face of the cube display the backscatter depth profile for a particular time period, in this case between 6:00 AM and 7:00 AM Pacific Standard Time on February 28th [1]. Expanding this in the Z dimension as a YZ plane shows how the vertical profile changes over calendar days. Similarly, a vertical cut parallel to the faces of the cube captures later days in the time series. The horizontal section represents the calendrical variation of daily events in depth.

These slices are relatively small subsets of a larger data set and can be exported from ENVI to Microsoft Excel, Matlab, or other programs for further analysis. Averaging data to reduce volume can be dangerous because it oversimplifies events and can be misleading. m The inset of is part of the highest temporal resolution data for February 28, showing important information about fish and zooplankton behavior. It is lost when averaging. This article uses black and white to display averaged data at full resolution and color to display time averages.

Example of a long-term series:

The seasonal period of the diurnal vertical profile Sv was calculated using a constant absorption coefficient of 0.05 dB m-1 and a sound speed of 1482 m s-1 based on the average groundwater temperature. (8.8℃) and salinity (30.8psu) for two years. Sv at noon (12:00 PST) was extracted from 2-min averaged data for 10 min, and these data were smoothed using a 5-point average to represent seasonal variations. From February to July in 2009-2010, a relatively thick layer of trash was observed near the ground during the day. This layer rose to a maximum depth of 70 meters in February 2009 and March 2010. A weak scattering layer with a more restricted vertical distribution was observed from June to September 2008 and from October 2009 to February 2010. Since the AWCP is shallower than the daily basic litter layer, it is not possible to determine the thickness of the litter layer [9].

Although a highly averaged view of the event, interpretation of the biological community underlying the signal is not possible. Although single-frequency sonar data theoretically does not support size segmentation of the signal, in practice fish and zooplankton separation can be inferred from full-resolution data based on texture and pattern interpretation. The full time resolution plot in a 1-hour vertical profile at 3 second intervals, demonstrating the richness of this data [3]. On May 1, 2009, the dispersion throughout the water column was low. Two weeks later, on May 15th, a large number of scattered shapes appeared 10 meters upstream, which may have been schools of herring, which are common in this area at this time of year.

Vertical Diel Movement Timing:

The strongest signal in the data is the Vertical Diel Movement (DVM) signal. Although the exact timing of the onset of the upward and downward movements is not always clear, the timing of the transition can be illustrated by cutting the data cube horizontally at intermediate depths. The left panel of the seasonal variation in the daily timing of the mean variance in the 39-41 m depth interval, as shown in the middle upper panel. The bottom center graphic shows the full moon graphic profile for each month in the left panel (bottom to top: March 12th, April 9th, May 9th, June 7th, July 7th) and moves It emphasizes the sharp and narrow peaks of scattering within. Observe zooplankton at a depth of 40m. The right panel of the timing of evening rise and dawn fall (defined here simply as the timing of maximum scattering at 40 meters during the dusk and dawn periods) relative to solar declination. and shows the highlights of its very close relationship with the length [2]. Variations in the timing of the DVM can be seen qualitatively in the left panel of “flare” or deviation from the sinusoid representing the evening rise and dawn fall. The simple "maximum scattering at depth" method for calculating DVM timing makes the calculations noisy and susceptible to strong signals, such as from fish. Fitting least squares at many depths for each day provides a more robust estimate. Cowardly. The time series of migration timing anomalies based on deviations from the solar declination relationship [10]. To assess whether these variations are due to false signals from the fish or true variations in DVM timing, we analyzed the full temporal resolution for the dates indicated by the colored circles.

Discussion:

As expected, analysis of the nascent long AWCP time series shows signs of a seasonal cycle. His two years of continuous acoustic observations in Saanich Inlet revealed seasonal changes in the litter layer during the day. The weak scattering layer regularly observed could be due to the lack of zooplankton or the depth of the scattering layer below the instrument. VENUS AWCP is located at a depth of 95m on the slope of a deep basin that reaches a maximum depth of 236m.

Repeated daytime acoustic surveys throughout the inlet using a downward-firing 200 kHz AWCP mounted on a ship have shown that deep scattering layers are generally located at depths of 75-125 m. Therefore, during the day part of the scattering layer is under the device. In Saanich Inlet, Eufausia pacifica is known to be the dominant species forming deep litter beds. Future analyzes will focus on nighttime profiles, where zooplankton is concentrated near the surface.

Diagram analysis strong scattering layer rising at dusk and descending at dawn in Saanich Inlet. This is consistent with observations of shorter time series by others. Even a very simple algorithm was able to detect when the scattering layer passes through an interval 40 m deep and showed a very strong correlation between timing and solar declination in time-averaged data.

However, using data with low temporal resolution or applying time averaging can introduce false anomalies. Examination of the high temporal resolution profiles both fish and physical phenomena in the form of internal waves can cause errors in signal interpretation. Smarter algorithms are needed that take advantage of the high temporal resolution of AWCP.

Even when instruments are configured to measure volumetric backscatter, fish targets are often observed throughout the water column, both as individual targets and as schools. Long time series are a rich source of behavioral data on both fish and zooplankton and their interactions. Fish targets are often seen mixing with and tracking temporally and spatially with more diffuse zooplankton clouds. This suggests that the fish feed on zooplankton. Reference did not have enough data to determine whether the fish descended with the zooplankton or whether the shoals were waiting near the bottom. You can see examples of both behaviors.

Conclusion:

The middle panel of is a particularly good example of a fish changing its swimming speed and direction to match the zooplankton, but continuing to migrate downward. Although data targeting fish have not yet been carefully analyzed, there are many examples of how the distribution of zooplankton affects fish, and where fish influence the distribution of zooplankton.

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